

Notes: Mossel, Sly, and Tamuz (Econometrica, 2015)

Strategic Learning and the Topology of Social Networks

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1 Big picture

- **Question.** When do strategic agents who repeatedly observe neighbors' actions eventually learn the true state of the world?
- **Setting.** Each agent receives one private signal about a binary state $S \in \{0, 1\}$ and then repeatedly chooses one of two actions. Agents observe past actions of their neighbors in a directed social network.
- **Strategic feature.** Agents are not myopic. They choose actions in a discounted repeated game, so current actions may be used to influence or extract future information.
- **Core topological condition.** The paper defines a network to be (d, L) -egalitarian if every agent observes at most d neighbors and if every observed edge has a return path of length at most L .
- **Main theorem.** On any connected countably infinite (d, L) -egalitarian network, every equilibrium leads to learning almost surely. On large finite connected (d, L) -egalitarian networks, learning occurs with probability arbitrarily close to one.
- **Why topology matters.** Bounded outdegree limits concentration of influence, and local return paths prevent one-way information hierarchies.
- **Agreement step.** In any equilibrium, all agents eventually agree on the same limiting action or set of actions. This builds on Rosenberg, Solan, and Vieille, plus nonatomic private beliefs.
- **Learning step.** Egalitarian topology creates many approximately independent points of view. If agents agree, independent estimators cannot all agree with high probability unless they identify the true state.
- **Failure examples.** In nonegalitarian networks, learning can fail. The royal family network violates local reciprocity; the mad king network violates bounded outdegree.
- **Technical contribution.** The paper builds a topology on rooted graph strategy profiles, proving compactness, equilibrium existence, and lower semicontinuity of learning probabilities.

2 Model and measurement

2.1 Information structure

- The state is binary: $S \in \{0, 1\}$, with prior $P[S = 0] = P[S = 1] = 1/2$.
- Each agent i receives a private signal W_i .
- Conditional on $S = 0$, signals are i.i.d. from μ_0 ; conditional on $S = 1$, signals are i.i.d. from μ_1 .
- The private belief is

$$I_i = P[S = 1 \mid W_i].$$

- The distribution of I_i is assumed nonatomic.

Interpretation: Nonatomic private beliefs rule out knife-edge indifference. This is crucial for turning agreement results into agreement on actual actions.

2.2 The social network

- The social network is a directed graph $G = (V, E)$.
- Agent i observes agents in

$$N(i) = \{j : (i, j) \in E\}.$$

- Each agent observes herself: $i \in N(i)$.

- A graph is strongly connected if there is a directed path between every ordered pair of vertices.
- A graph is L -locally connected if whenever $(i, j) \in E$, there is a path from j back to i of length at most L .
- A graph is (d, L) -egalitarian if all outdegrees are bounded by d and the graph is L -locally connected.

Interpretation: Egalitarianism is a geometric restriction on influence. No agent can observe too many others, and no agent is too far removed from those who observe her.

2.3 Repeated game

- Time is discrete: $t = 0, 1, 2, \dots$
- At each time, each agent chooses an action $A_i^t \in \{0, 1\}$.
- A pure strategy at time t maps the agent's private belief and neighbors' past actions into an action.
- The action under strategy profile Q is

$$A_i^t = A_i^t(G, Q) = Q_i^t(I_i, A_{0,t}^{N(i)}).$$

Interpretation: Actions can be strategic signals. An agent may care not only about current payoff but also about how current action affects future learning by others and herself.

2.4 Payoffs and equilibrium

The stage utility is

$$U_{i,t}(G, Q) = \mathbf{1}\{A_i^t(G, Q) = S\}.$$

Expected stage utility is

$$u_{i,t}(G, Q) = P[A_i^t(G, Q) = S].$$

With discount factor $\lambda \in (0, 1)$,

$$u_i(G, Q) = (1 - \lambda) \sum_{t=0}^{\infty} \lambda^t u_{i,t}(G, Q).$$

A strategy profile Q is a Nash equilibrium if no agent can improve u_i by deviating while all other agents keep their strategies fixed.

Interpretation: The normalization $(1 - \lambda)$ keeps lifetime utility in $[0, 1]$. Discounting is also important technically because faraway agents' strategies have diminishing influence on current utility.

3 Models and key theoretical objects

3.1 Learning

Agent i learns the state if from some finite time onward she always chooses the true action:

$$\exists T < \infty \text{ such that } A_i^t = S \quad \forall t \geq T.$$

Learning takes place if every agent learns S .

Interpretation: Learning is stronger than high posterior belief at one date. It means eventual correct behavior forever.

3.2 Agreement

Define the infinite action set of agent i :

$$C_i = \{s \in \{0, 1\} : A_i^t = s \text{ for infinitely many } t\}.$$

Theorem 5.1. In any equilibrium, with probability one,

$$C_i = C_j \quad \text{for all agents } i, j.$$

Interpretation: Strategic agents eventually agree in the sense that their long-run action sets coincide. This does not yet prove learning: everyone could agree on the wrong action.

3.3 Best responses and information at infinity

Let $\mathcal{F}_i^t$ denote agent i 's information at time t , and let $\mathcal{F}_i^\infty$ be the limiting information. The myopic best response is

$$B_i^t = \arg \max_{s \in \{0, 1\}} P[s = S \mid \mathcal{F}_i^t].$$

The limiting best-response set is

$$B_i^\infty = \arg \max_{s \in \{0, 1\}} P[s = S \mid \mathcal{F}_i^\infty].$$

Interpretation: Agreement is proven by combining game-theoretic agreement results with the fact that nonatomic beliefs remove indifference at finite times.

3.4 Rooted graph topology

A rooted graph (G, i) is a graph viewed from the perspective of root i . For two rooted graphs, the distance is

$$D([G, i], [G', i']) = \inf\{2^{-r} : B_r(G, i) \cong B_r(G', i')\}.$$

Interpretation: Two rooted graphs are close if large neighborhoods around their roots look the same after relabeling vertices.

3.5 Compactness and egalitarianism

Let $\mathcal{R}(G)$ be the set of rooted views $[G, i]$ generated by all vertices i in G .

Lemma 6.1. For a strongly connected graph G , the closure of $\mathcal{R}(G)$ is compact if and only if G is (d, L) -egalitarian for some d and L .

Interpretation: Egalitarian graphs are exactly those with finitely many approximate points of view. This compactness property powers the main learning argument.

3.6 Rooted graph strategy profiles

A rooted graph strategy profile $[G, i, Q]$ is a network, root, and equilibrium strategy profile as seen from the root. The paper defines a metric combining local graph similarity and local strategy similarity.

Important properties:

- Expected utility $u([G, i, Q])$ is continuous.
- The set of equilibrium rooted graph strategy profiles is closed.
- The probability of learning map $p([G, i, Q])$ is lower semicontinuous.
- On egalitarian graphs, the space of rooted graph strategy profiles is precompact.

Interpretation: These topological tools are needed because the equilibrium can be nonsymmetric and the network can be infinite.

4 Identification of the mechanism

4.1 Why symmetric-grid intuition works

In the two-dimensional grid, choose two agents far enough apart that information cannot travel between them by time T . Conditional on the state, their actions at time T are independent. But agreement says they must converge to the same action with high probability. Independent variables that agree with high probability must be nearly constant, and because private signals are informative, the common limiting action must be the true state.

Interpretation: Agreement plus independence implies learning. Egalitarian topology generalizes this intuition beyond symmetric grids.

4.2 Why compactness replaces symmetry

In nonsymmetric egalitarian graphs, agents need not be identical. Compactness says that there are only finitely many approximate types of viewpoints. This permits the proof to find many agents with nearly the same point of view, separate them far apart, and create approximately independent estimators of S .

Interpretation: Compactness is the formal substitute for symmetry.

4.3 Why nonegalitarian networks fail

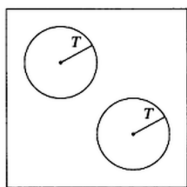
- **Royal family.** A fixed small elite is observed by a large public. If the elite is initially wrong, the public can imitate the wrong action with probability bounded away from zero.
- **Mad king.** The graph is undirected but has unbounded degree at the king. Strategic threats can cause the people to suppress their information, so their private signals are never aggregated.

Interpretation: Each counterexample violates one side of egalitarianism: the royal family violates local return paths, while the mad king violates bounded outdegree.

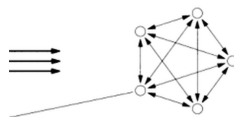
5 Tables, figures, and original equation panels

The paper has no empirical tables. The visual material consists of network diagrams and original mathematical panels. The panels below are directly cropped from the paper and grouped compactly.

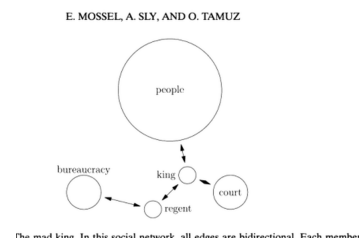
5.1 Figures 1–3: Learning and nonlearning networks



(a) Figure 1: grid intuition.



(b) Figure 2: royal family.



(c) Figure 3: mad king.

Read:

- Figure 1 illustrates the proof idea on the two-dimensional grid: faraway agents' actions are independent by a finite time but must eventually agree.
- Figure 2 shows a nonegalitarian hierarchy: a fixed royal family can mislead a large public.
- Figure 3 shows a nonegalitarian high-degree structure: the king becomes a strategic bottleneck for information aggregation.

Economic message: Egalitarian networks allow many independent estimators to accumulate. Nonegalitarian networks can create information bottlenecks or strategic suppression of private signals.

5.2 Panel 1: Information structure and graph primitives

ow half. To avoid encumbering this paper with an additional layer of technical complexity, we focus on the case in which $\mathbb{P}[S = 1] = \mathbb{P}[S = 0] = 1/2$. Agent i 's *private belief* I_i is the probability that the state of the world is 1, given i 's private signal:

$$I_i = \mathbb{P}[S = 1 | W_i].$$

ce I_i is a sufficient statistic for S given W_i , we assume below without loss of generality that an agent's actions depend on W_i only through I_i (see, e.g., Smith and Sørensen (2000)).

We consider only μ_0 and μ_1 such that the distribution of I_i is nonatomic. This is the condition that we refer to above as *nonatomic private beliefs*. This is an additional restriction that we impose, beyond what is standard in the herding literature.

4.2. The Social Network

The agents' social network defines which of them observe the actions of which others. We do not assume that this is a symmetric relation: it may be that i observes j while j does not observe i . Formally, the social network $G = (V, E)$ is a directed graph: V is the set of agents and E is a relation on V or a subset of the set of ordered pairs $V \times V$. The set of neighbors of $i \in V$ is

$$N(i) = \{j : (i, j) \in E\},$$

and we consider only graphs in which $i \in N(i)$; that is, we require that an agent observes her own actions. The *out-degree* of i is given by $|N(i)|$ and will always be finite. This means that an agent observes the actions of a finite number of other agents. We do allow infinite *in-degrees*; this corresponds to agents whose actions are observed by infinitely many other agents.

Let $G = (V, E)$ be a directed graph. A (directed) *path* of length k from $i \in V$ to $j \in V$ in G is a sequence of $k + 1$ nodes $i_1, \dots, i_{k+1}$ such that $(i_n, i_{n+1}) \in E$ for $n = 1, \dots, k$, and where $i_1 = i$ and $i_{k+1} = j$.

Figure 2: Original panel: private beliefs, neighbor sets, and graph definitions.

Read:

- The private belief $I_i = P[S = 1 | W_i]$ is the sufficient statistic for an agent's signal.
- The neighbor set $N(i)$ determines whose actions agent i observes.
- (d, L) -egalitarianism combines bounded outdegree with local return paths.

Economic message: The model isolates how topology controls the flow of strategic information.

5.3 Panel 2: Strategies, payoffs, and equilibrium

available to i at time t is her own private signal (of which the relevant information is her private belief, taking values in $[0, 1]$) and the actions of her neighbors in previous time periods, taking values in $\{0, 1\}^{|N(i)|:t}$. This action is hence calculated by some function from $[0, 1] \times \{0, 1\}^{|N(i)|:t}$ to $\{0, 1\}$.

A *pure strategy* at time t of an agent $i \in V$ is therefore a Borel-measurable action $q_t^i: [0, 1] \times \{0, 1\}^{|N(i)|:t} \rightarrow \{0, 1\}$. A *pure strategy* of an agent i is the sequence of functions $q^i = (q_0^i, q_1^i, \dots)$, where q_t^i is i 's pure strategy at time t . We endow the space of pure strategies with the topology derived from the weak topology on functions from $[0, 1]$ to $\{0, 1\}$.

A *mixed strategy* Q^i of agent i is a pure-strategy-valued random variable; this is the standard notion of a mixed strategy, and we shall henceforth refer to mixed strategies simply as strategies. A (mixed) *strategy profile* is a set of strategies $\bar{Q} = \{Q^i: i \in V\}$, where the random variables Q^i are independent of each other and of the private signals.

The *action* of agent i at time t is denoted by $A_t^i \in \{0, 1\}$. Denote the *history* actions of the neighbors of i before time t by $A_{0:t}^{N(i)} = \{A_s^j: s < t, j \in N(i)\}$; this depends on the social network G . The action that agent i plays at time t under strategy profile $\bar{Q}$ is

$$A_t^i = A_t^i(G, \bar{Q}) = Q_t^i(I_i, A_{0:t}^{N(i)}).$$

Note again that we (without loss of generality) limit the action to be a function of the private belief I_i , as opposed to the private signal W_i .

(a) Actions and strategies.

Let $0 < \lambda < 1$ denote the agents' common *discount factor*. Given a social network G and strategy profile $\bar{Q}$, agent i 's *stage utility* at time t , $U_{i,t}$, is 1 if her action matches S and is 0 otherwise:

$$U_{i,t} = U_{i,t}(G, \bar{Q}) = \mathbf{1}_{A_t^i(G, \bar{Q})=S}.$$

agent i 's *expected stage utility* at time t , $u_{i,t}$, is therefore given by

$$u_{i,t} = u_{i,t}(G, \bar{Q}) = \mathbb{E}[U_{i,t}(G, \bar{Q})] = \mathbb{P}[A_t^i(G, \bar{Q}) = S].$$

agent i 's *expected utility* u_i is given by

$$u_i = u_i(G, \bar{Q}) = (1 - \lambda) \sum_{t=0}^{\infty} \lambda^t u_{i,t}(G, \bar{Q}).$$

Note that $u_i \in [0, 1]$ due to the normalization factor $(1 - \lambda)$. A *game* $\mathcal{G}$ is a tuple $(\mu_0, \mu_1, \lambda, G)$ consisting of two measures, a discount factor, and a social network graph, that satisfy the conditions of the definitions above.

4.4. Equilibria

A *Bayesian equilibrium* concept is the standard Nash equilibrium in games of incomplete information: $\bar{Q}$ is an equilibrium if no agent can improve her expected utility $u_i(\bar{Q})$ by deviating from $\bar{Q}$.

Formally, in a game $\mathcal{G} = (\mu_0, \mu_1, \lambda, G)$, strategy profile $\bar{Q}$ is an equilibrium if for every agent $i \in V$, it holds that

(b) Utilities and equilibrium.

Read:

- A pure strategy maps private belief and observed action histories into an action.
- Stage utility equals one when the action matches the state.
- Lifetime utility is discounted and normalized.
- Equilibrium is Nash equilibrium in a discounted game of incomplete information.

Economic message: The strategic problem is richer than myopic learning because each action can shape later information flows.

5.4 Panel 3: Agreement and best responses

THEOREM 5.1: *Let $\mathcal{G}$ be a game with either finitely many players or countably infinitely many players, and let $\bar{Q}$ be an equilibrium strategy profile of $\mathcal{G}$. Then, with probability 1, $C_i = C_j$ for all agents $i, j \in V$.*

This theorem is a crucial ingredient in the proof of the main result of this article. Indeed, learning occurs if $C_i = S$ for all i , and so a prerequisite is that $C_i = C_j$ for all i, j .

Recall that a strategy of agent i at time t is a function of her private belief and the actions of her neighbors in previous time periods, $A_{[0,t)}^{N(i)}$. Hence we think of the sigma-algebra generated by these random variables as the “information available to agent i at time t .” Denote the information available to agent i at time t by

$$\mathcal{F}_t^i = \mathcal{F}_t^i(G, \bar{Q}) = \sigma(I_i, Q^i, A_{[0,t)}^{N(i)})$$

and denote by

$$\mathcal{F}_\infty^i = \mathcal{F}_\infty^i(G, \bar{Q}) = \sigma\left(\bigcup_{t=0}^{\infty} \mathcal{F}_t^i\right)$$

the information available to agent i at the limit $t \rightarrow \infty$. Note that $\mathcal{F}_t^i$ includes the sigma-algebra generated by i ’s private belief, the actions of i ’s neighbors before time t , and i ’s pure strategy; i knows which pure strategy she has chosen. Since the expected stage utility of action s at time t is $\mathbb{P}[s = S]$, a myopic agent would take an action s in $\{0, 1\}$ that maximizes $\mathbb{P}[s = S | \mathcal{F}_t^i]$. This motivates the following definition. Denote the best response of agent i at time t

$$B_t^i = B_t^i(G, \bar{Q}) = \arg \max_{s \in \{0,1\}} \mathbb{P}[s = S | \mathcal{F}_t^i(G, \bar{Q})].$$

likewise denote the set of best responses of agent i at the limit $t \rightarrow \infty$ by

$$B_\infty^i = B_\infty^i(G, \bar{Q}) = \arg \max_{s \in \{0,1\}} \mathbb{P}[s = S | \mathcal{F}_\infty^i].$$

Figure 4: Original panel: agreement theorem, information sigma-algebras, and best responses.

Read:

- Theorem 5.1 says all agents eventually have the same infinite action set.
- The proof uses information at finite times and at the limit.
- Best responses are defined using posterior probabilities conditional on those information sets.

Economic message: Agreement is necessary but not sufficient for learning. The remaining step is to show that the common limit equals the true state.

5.5 Panel 4: Rooted graph compactness

rooted graph (G, i) is a rooted graph, $(G, i) \neq (G', i')$, since the graph is directed. The (directed) ball $B_r(G, i)$ of radius r of the rooted graph (G, i) is the rooted graph, with root i , induced in G by the set of vertices $\{j \in V : d(i, j) \leq r\}$.

We now proceed to define our topology on the space of isomorphism classes of strongly connected rooted graphs, which is an extension of the Benjamini-Schramm (2011) topology on undirected graphs. We define this topology as a metric.⁷

Let $[G', i']$ and $[G, i]$ be isomorphism classes of strongly connected rooted graphs. The distance $D([G, i], [G', i'])$ is defined by

$$D([G, i], [G', i']) = \inf\{2^{-r} : B_r(G, i) \cong B_r(G', i')\}.$$

That is, the larger is the radius around the roots in which the graphs are isomorphic, the closer they are. In fact, the quantitative dependence of $D(\cdot, \cdot)$ on r (exponential in our definition) will not be of importance here, as we shall only be interested in the topology induced by this metric.

It is straightforward to show that $D(\cdot, \cdot)$ is well defined; a standard diagonalization argument (which we use repeatedly in this article) is needed to show that it is indeed a metric rather than a pseudometric (Claim A.1). The assumption of strong connectivity is crucial here, since $D(\cdot, \cdot)$ is otherwise a pseudometric.

Let SCG be the set of isomorphism classes of *strongly connected* rooted graphs. This set is a topological space when equipped with the topology induced by the metric $D(\cdot, \cdot)$. Given a strongly connected graph G , let $\mathcal{R}(G) \subset SCG$ be the set of all rooted graph isomorphism classes of the form $[G, i]$ for i a vertex in G . This can be thought of as the set of points of view in the graph G . The notion of (d, L) -egalitarianism now arises naturally in the sense that the number of approximate points of view in G is finite if and only if G is egalitarian. This is formalized in the following lemma.

LEMMA 6.1: *Let G be a strongly connected graph. Then the closure of $\mathcal{R}(G)$ is compact in SCG if and only if G is (d, L) -egalitarian for some d and L .*

We would like to suggest that Lemma 6.1, which we prove in Appendix A,

Figure 5: Original panel: rooted graph distance and compactness lemma.

Read:

- Two rooted graphs are close when large balls around the roots are isomorphic.
- Lemma 6.1 links compactness of rooted views to (d, L) -egalitarianism.
- This topological characterization is the technical bridge from network geometry to learning.

Economic message: The graph is learnable because it has no infinitely many radically different local vantage points.

5.6 Panel 5: Main learning theorem

THEOREM 7.1: *Fix the distributions of the agents' private signals with nonatomic beliefs. Fix also a discount factor $\lambda \in (0, 1)$, and positive integers L and d . Then in any connected, (d, L) -egalitarian, countably infinite network,*

$$\mathbb{P}[\text{all agents learn } S] = 1$$

in any equilibrium. Furthermore, for every $\varepsilon > 0$, there exists an n such that for any connected, (d, L) -egalitarian network with at least n agents,

$$\mathbb{P}[\text{all agents learn } S] \geq 1 - \varepsilon$$

in any equilibrium.

Given a set of graphs $\mathcal{K}$, let $\mathcal{R}(\mathcal{K})$ be the set of rooted graphs $[G, i]$ such that $[G, i] \in \mathcal{K}$. Let $\mathcal{EQ}(\mathcal{K})$ be the set of equilibrium strategy profiles $[G, i, Q]$ such that $[G, i] \in \mathcal{K}$.

PROOF OF THEOREM 7.1: Let G be a (d, L) -egalitarian graph. The case that G is finite is treated in Theorem 7.1.

Figure 6: Original panel: Theorem 7.1.

Read:

- Infinite connected (d, L) -egalitarian networks learn almost surely in any equilibrium.
- Large finite connected (d, L) -egalitarian networks learn with probability at least $1 - \varepsilon$.
- The result holds for any fixed discount factor and signal distributions with nonatomic private beliefs.

Economic message: The result is robust across equilibria: topology alone ensures strategic information aggregation under egalitarianism.

6 Short summary

- The paper studies social learning by fully strategic, forward-looking agents on directed networks.
- Agents repeatedly choose between two actions and observe neighbors' past actions.
- The true state is binary and private signals are conditionally independent.
- The key network condition is (d, L) -egalitarianism: bounded outdegree plus bounded return paths.
- In any equilibrium, all agents eventually agree on the same limiting action set.
- On infinite egalitarian networks, agreement plus compactness implies almost sure learning.
- On large finite egalitarian networks, learning occurs with probability arbitrarily close to one.
- Nonegalitarian networks can fail to aggregate information because of hierarchy, bottlenecks, or strategic suppression.
- The paper develops a topology on rooted graph strategy profiles to handle nonsymmetric equilibria and infinite networks.
- The main lesson is that strategic learning depends on network topology: egalitarian networks aggregate information, nonegalitarian networks may not.

7 Conclusion

7.1 Core conclusion

Mossel, Sly, and Tamuz show that topology determines whether strategic social learning succeeds. In connected egalitarian networks, every equilibrium aggregates private information, so agents eventually learn the true state. In nonegalitarian networks, learning can fail even when agents are rational and the network is strongly connected.

7.2 Economic intuition

Learning requires both agreement and independent information. Strategic agents eventually agree, but agreement could be wrong. Egalitarian topology prevents information from being concentrated in a few agents or trapped in one-way hierarchies. It creates many approximately independent local viewpoints, forcing the common limiting action to identify the true state.

7.3 Takeaway

The paper provides a strategic foundation for why egalitarian social networks support information aggregation. The important restriction is not simply connectedness; it is the absence of severe influence asymmetries. Networks with elites, bottlenecks, or unbounded influence can block learning even when everyone is rational.